Department of Information Systems and Technologies

CTIS221-Object Oriented Programming

FALL 2026-2027

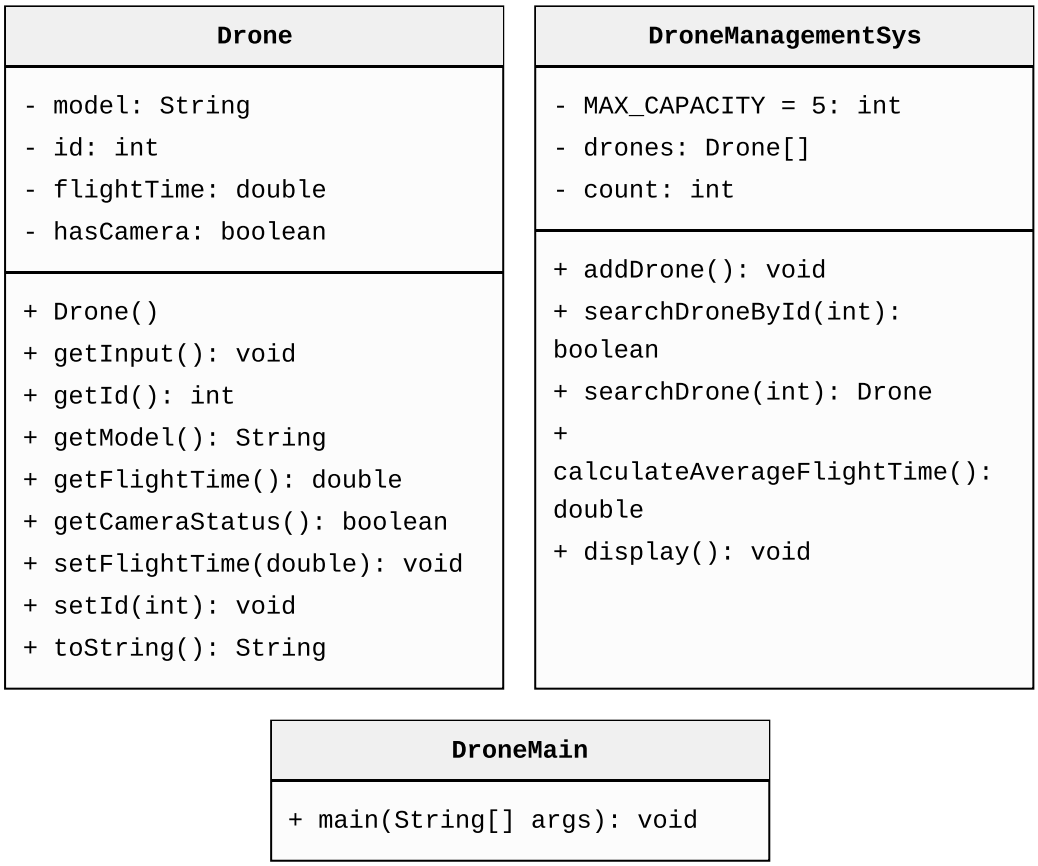
Lab Guide #9 (Practice)

OBJECTIVES: Object Methods, Array of objects, System Classes
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Q1. Write a Java program that manages and displays information of `Drone` objects. Create a `Drone` class structure with the given data members. Also, implement a no-arg constructor, necessary getters & setters, and a `toString()` method.

Then create a `DroneManagementSys` class that has a static data member of type int named `MAX_CAPACITY` (constant), a static data member of type `Drone[]` named `drones`, and a static integer `count` data member. Also, implement the static methods `addDrone()`, `searchDroneById()`, `searchDrone()`, `calculateAverageFlightTime()`, and `display()`.

Finally, create a `DroneMain` class that has the static `main()` method.



Create a `Drone` class with the following instructions:

- Implement a `getInput()` method that gets the data member values from the user. The method should ask the user if the drone has a camera, taking (Y/y or N/n) as input and setting the boolean value of `hasCamera` accordingly.
- Implement necessary getter & setter methods.
- Implement a `toString()` method. The method should check whether the drone has a camera or not. Display the camera status accordingly.

Create a `DroneManagementSys` class with the following instructions:

- Declare a static final member `MAX_CAPACITY` to define the maximum size of the drones array. The value will be 5.
- Declare and create an object array `drones` of type `Drone[]` with the `MAX_CAPACITY` constant value (5) as its size.
- Declare a static member `count` to store the current number of drones.
- Implement an `addDrone()` method that gets drone information by invoking the `getInput()` method for each Drone object. It should check whether a drone ID is already assigned by invoking the `searchDroneById()` method and store the object in the `drones` array if the ID is available. Also, ask the user if they want to continue adding new drones ((Y/N) Y for yes, N for no). If the user enters 'N', the loop terminates. The method also checks whether `MAX_CAPACITY` has been reached or not.
- Implement a `searchDroneById()` method that takes a drone ID as a parameter, searches the ID in each Drone object in the `drones` array, and returns `true` if the ID is available, otherwise `false`.
- Implement a `searchDrone()` method that takes the id of the drone as a parameter and searches for the drone id in the `drones` array. If found, return the object; else, return `null`.
- Implement a `display()` method to display the objects in the Drones array.
- Implement a `calculateAverageFlightTime()` method in the `DroneManagementSys` class that loops through the Drones array and calculates the average flight time of all drones in the array. It should return the average flight time value as a double. If no drones are present, it should return 0.

Create a `DroneMain` class with the following instructions:

- Input 5 Drone objects by invoking the `addDrone()` method of the `DroneManagementSys` class.
- Try to input a Drone object that has the same drone ID as one of the drone objects already in the array.
- Try to add 2 more Drone objects to exceed `MAX_CAPACITY`.
- Display all drones by invoking the `display()` method of the `DroneManagementSys` class.

- Calculate the average flight time of drones in the array by invoking the `calculateAverageFlightTime()` method, and display the result.
- Ask the user which drone to search by ID, and then display it using the `searchDrone()` method. Continue prompting until a valid drone is found.

Output:

```
Enter Drone Id: 1 Enter Drone Model: DJI Mavic 3 Enter Drone Flight Time: 45,5 Does drone have a camera? (Y/N): Y Drone added successfully! Do you want to add another drone? (Y/N): Y Enter Drone Id: 2 Enter Drone Model: FPV Custom Enter Drone Flight Time: 12,0 Does drone have a camera? (Y/N): N Drone added successfully! Do you want to add another drone? (Y/N): Y Enter Drone Id: 2 Drone ID already exists. Please enter a different ID. Enter Drone Id: 3 Enter Drone Model: Autel Evo Enter Drone Flight Time: 35,2 Does drone have a camera? (Y/N): Y Drone added successfully! Do you want to add another drone? (Y/N): N ----------------- Here is a list of the Drones: Drone Model=DJI Mavic 3, Id=1, Flight Time=45.5, Camera Status=Drone has a camera. Drone Model=FPV Custom, Id=2, Flight Time=12.0, Camera Status=No camera. Drone Model=Autel Evo, Id=3, Flight Time=35.2, Camera Status=Drone has a camera. ---------------- Avg of all Drone Flight Times: 30,90 ----------------- Which drone would you like to search? Enter its id: 4 There is no such drone in the array Which drone would you like to search? Enter its id: 1 Here is the drone you were looking for... Drone Model=DJI Mavic 3, Id=1, Flight Time=45.5, Camera Status=Drone has a camera.
```